



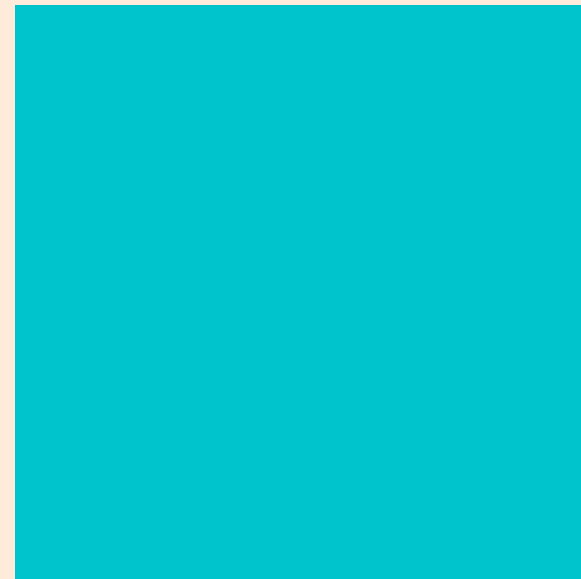
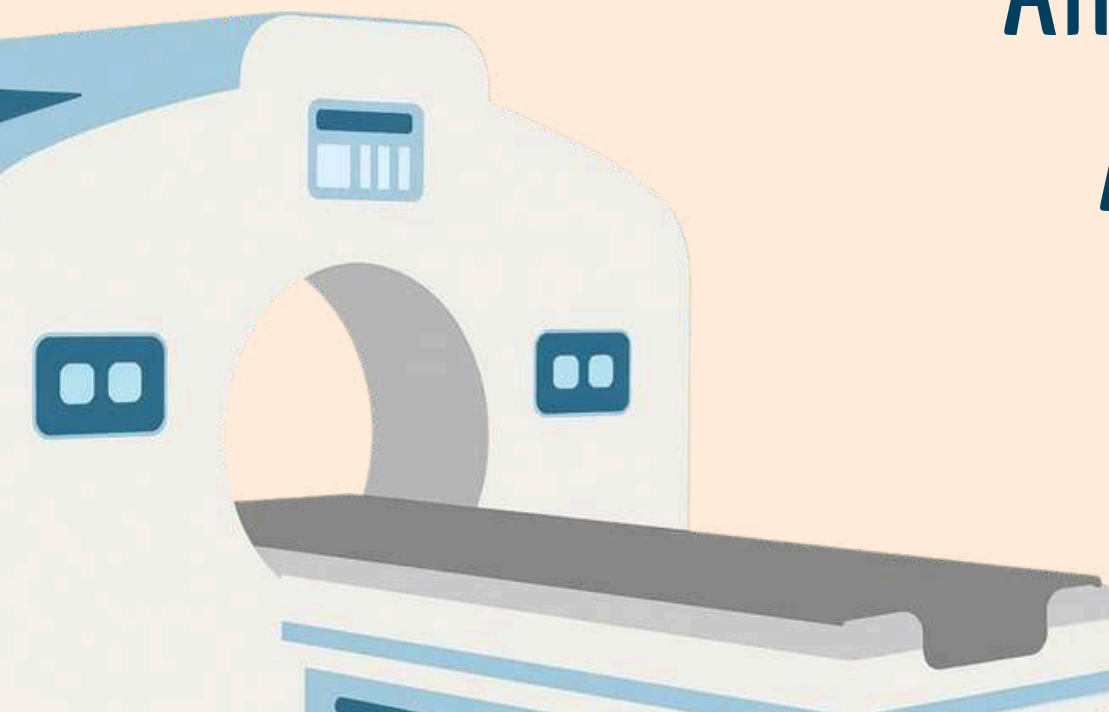
Lung SynthAI

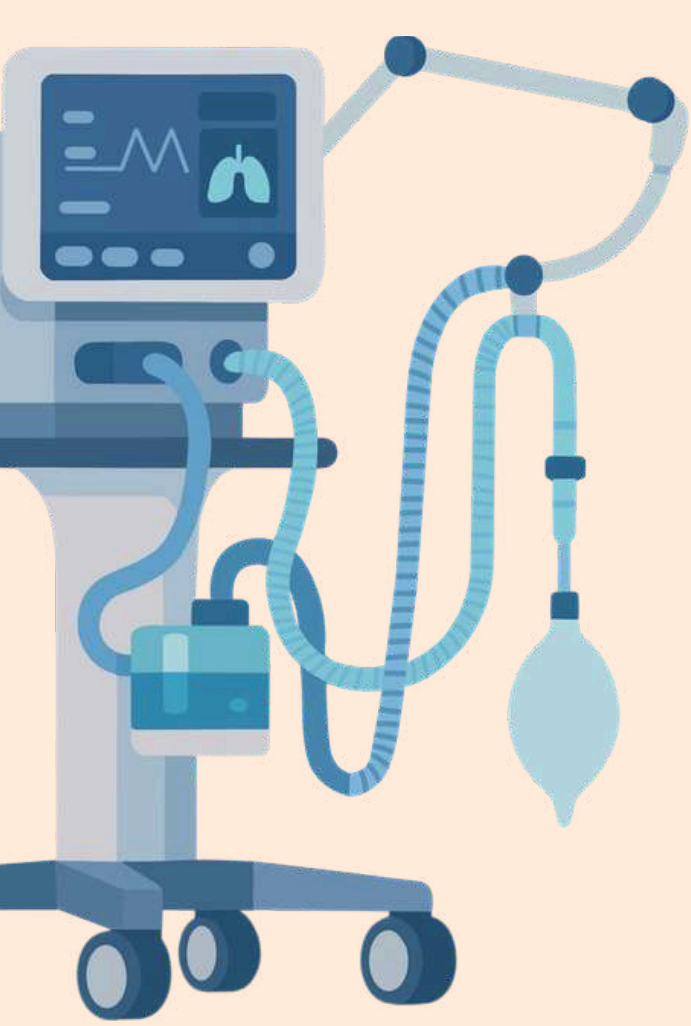
**“Phase-Conditioned
Diffusion Model for 4D Lung CT Intermediate
Phase Reconstruction”**

Ahnaf Mohammed Mahi Kabir (2222171042)

Afia Mubassira Ali Raisa (2121812642)

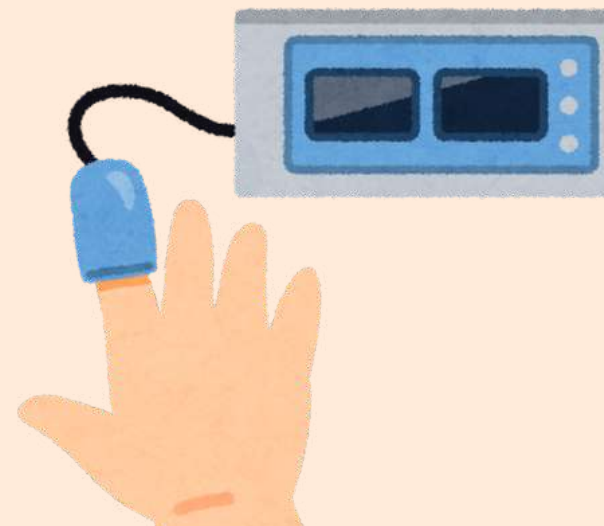
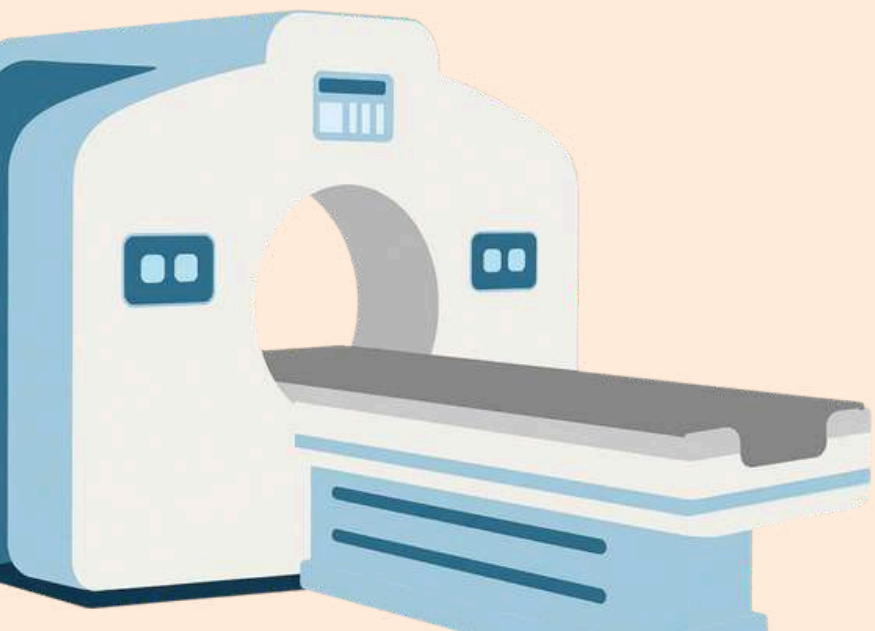
Samiha Tasnim Diba (2212515642)



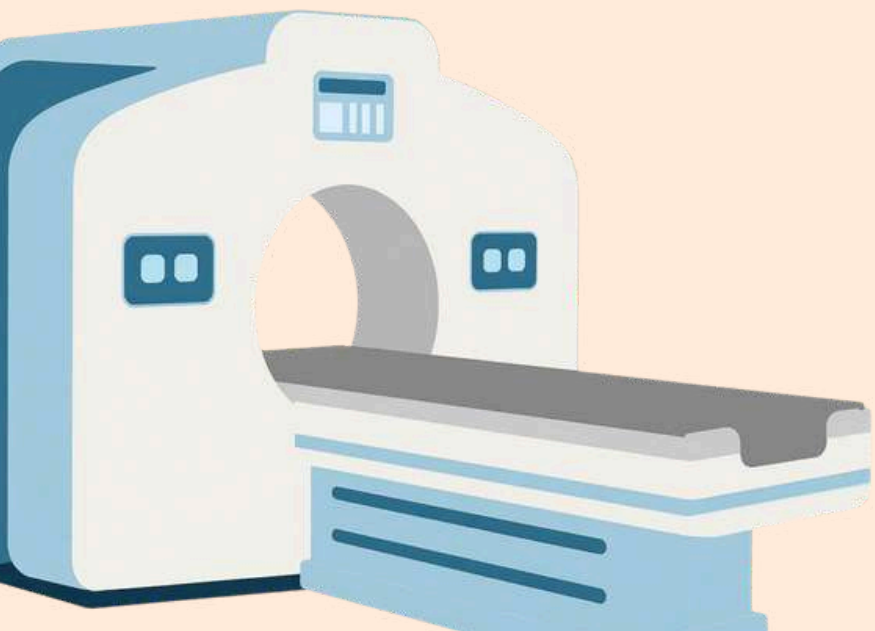
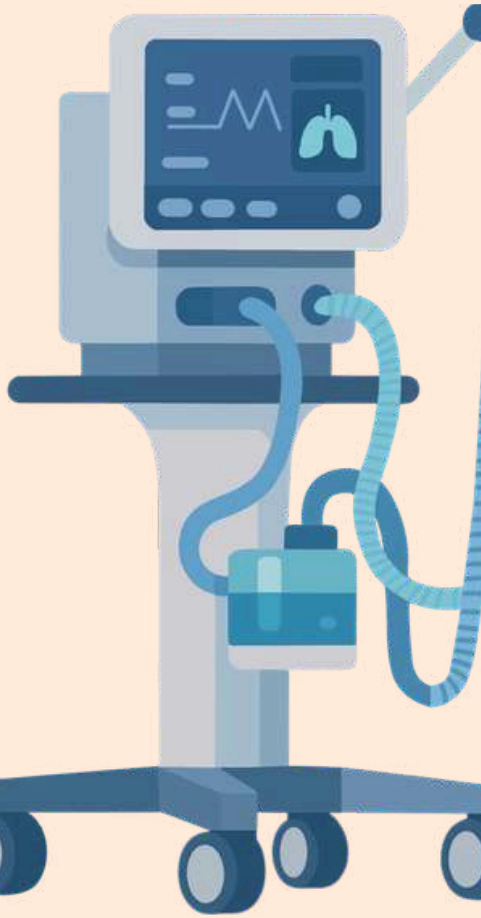
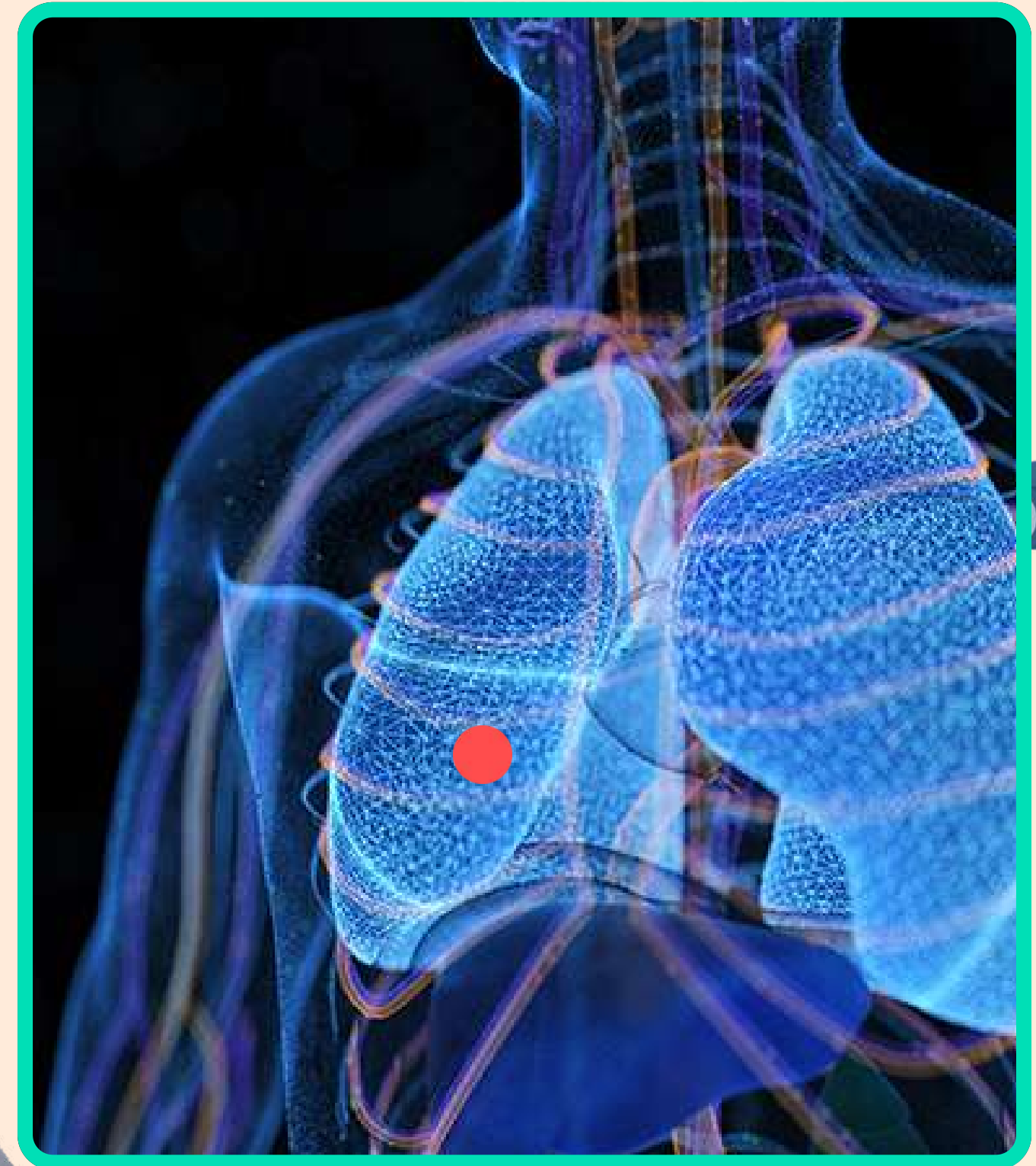
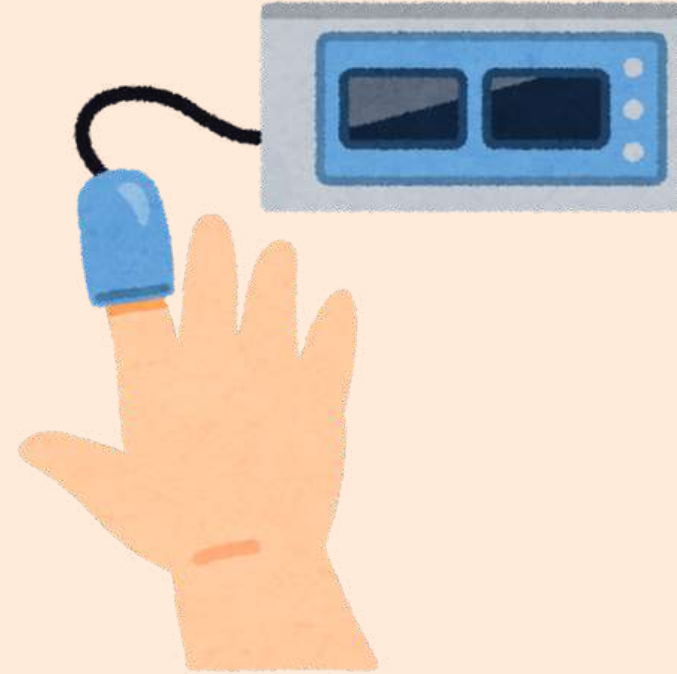


Focus of the project

- Problem during radiotherapy
- 10 phases are captured
- Fully inhaled & fully exhaled phases are the clearest

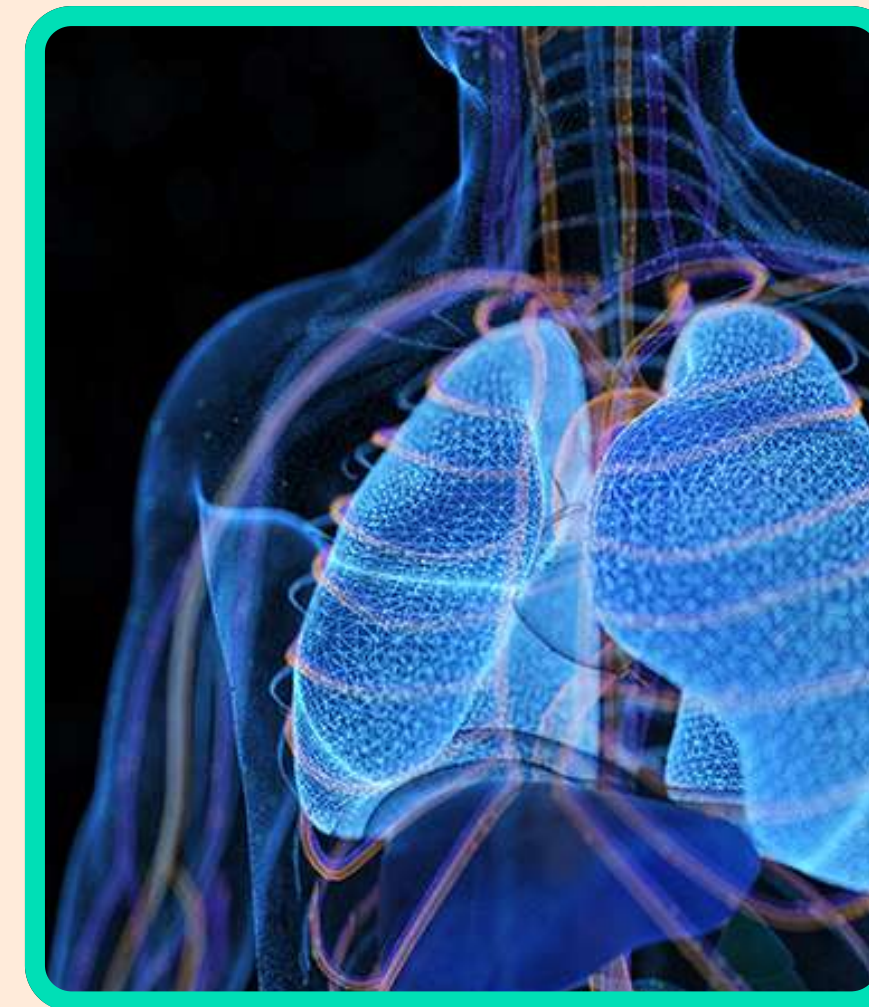
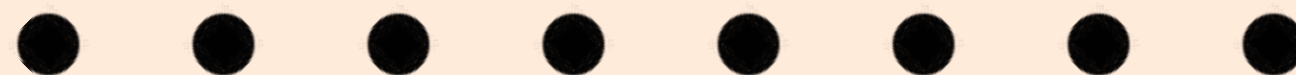


- Intermediate phases can be blurry due to lung movement
- Healthy tissue exposed to radiation

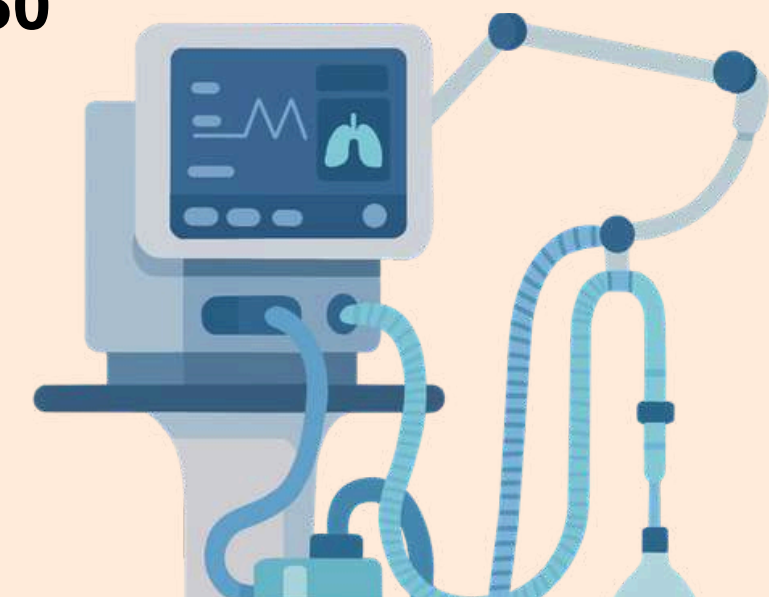
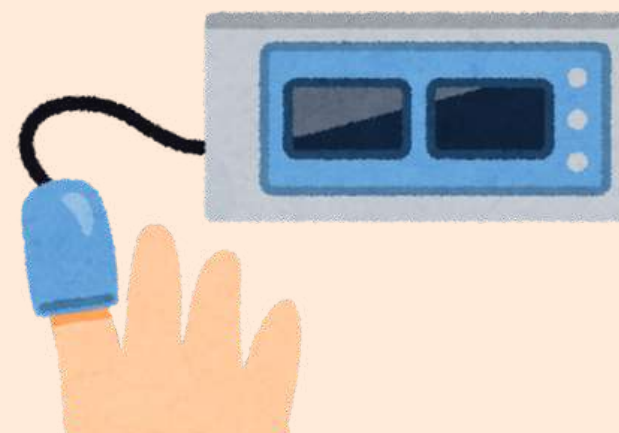




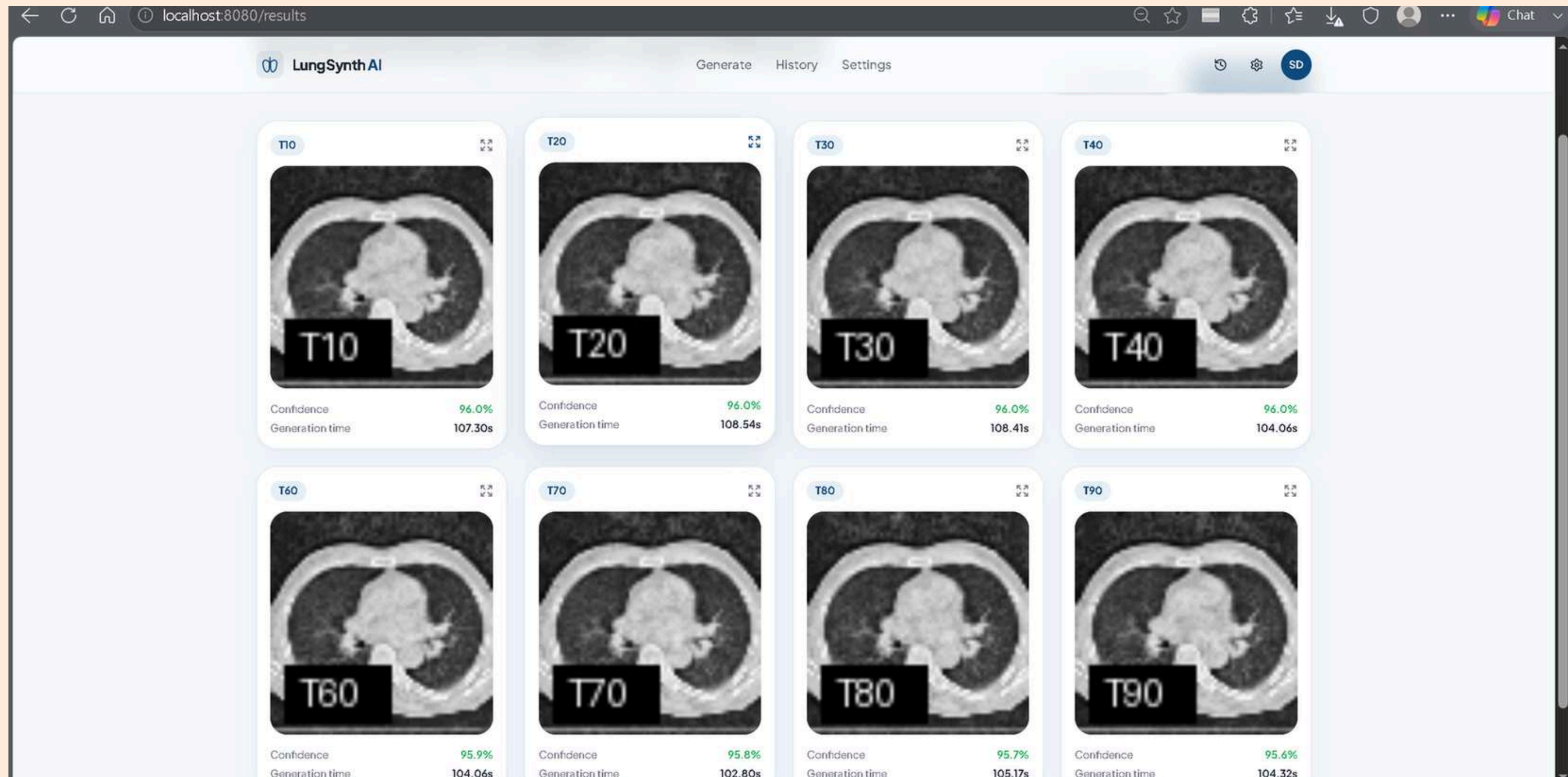
T00

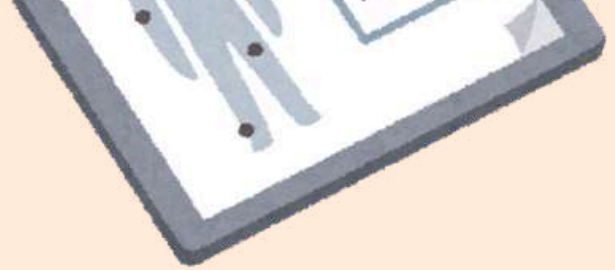


T50



Intermediate Breathing Phases

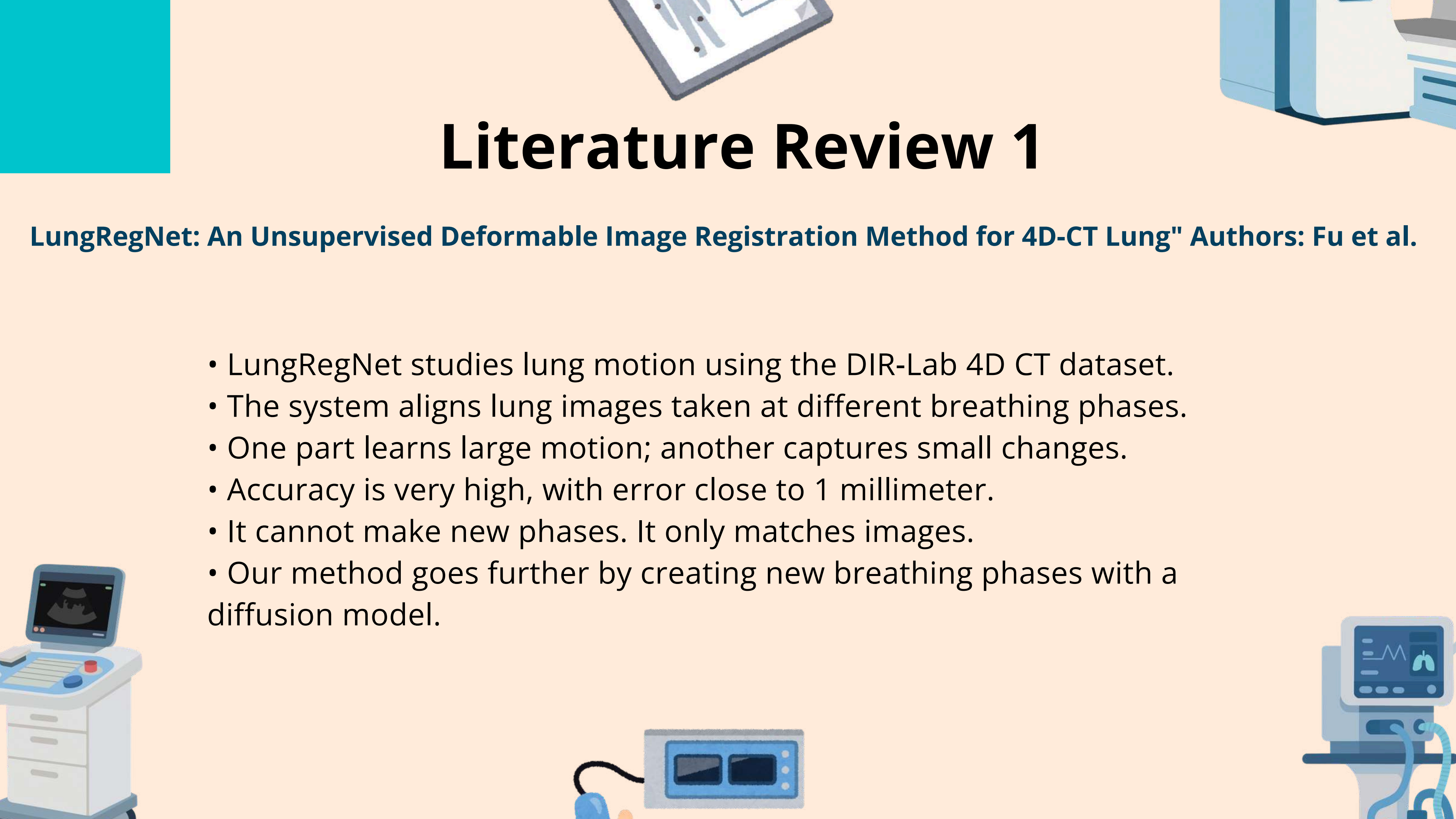




DIR - Lab 4D-CT dataset

- Contains a small number of unique patients
- Dataset Expansion Strategy
- Generating synthetic training data using "frontier models"
- Failed to generate 4D images (sensitive tissue information)
- Lack of Medical Precision and Voxel Accuracy

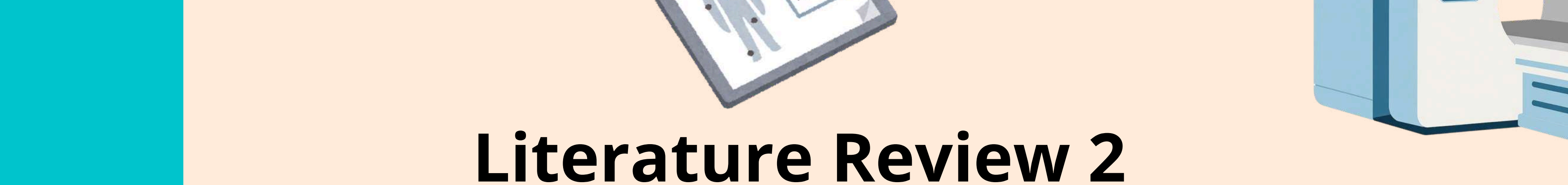




Literature Review 1

LungRegNet: An Unsupervised Deformable Image Registration Method for 4D-CT Lung" Authors: Fu et al.

- LungRegNet studies lung motion using the DIR-Lab 4D CT dataset.
- The system aligns lung images taken at different breathing phases.
- One part learns large motion; another captures small changes.
- Accuracy is very high, with error close to 1 millimeter.
- It cannot make new phases. It only matches images.
- Our method goes further by creating new breathing phases with a diffusion model.

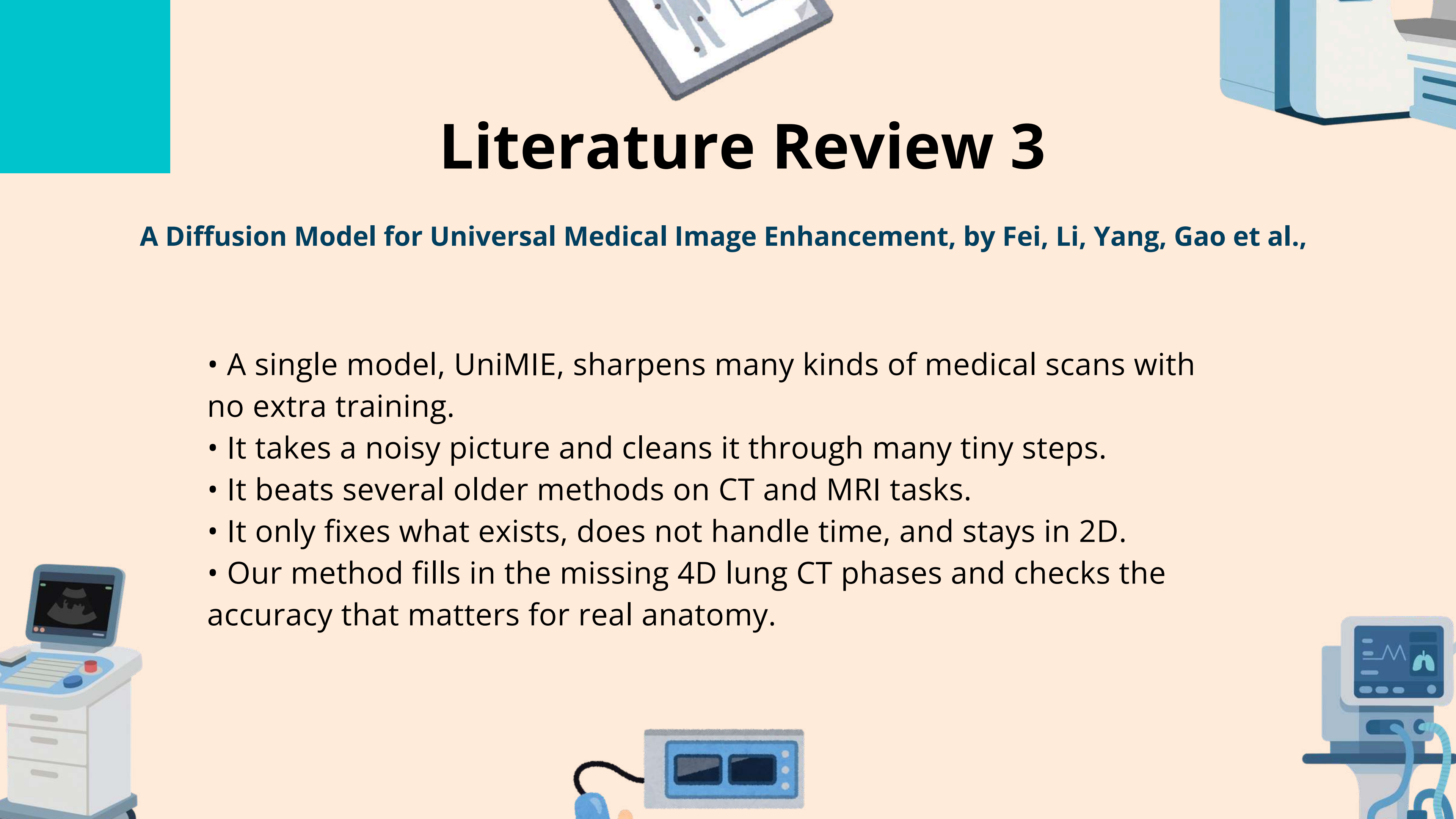


Literature Review 2

Fast-DDPM: Fast Denoising Diffusion Probabilistic Models for Medical Image-to-Image Generation" Authors: Jiang et al.

- Fast-DDPM proves diffusion models are powerful tools for medical imaging.
- Tested on three tasks: prostate MRI enhancement, CT denoising, and MRI-to-MRI conversion.
- Achieves better results than older approaches such as CNNs and GANs.
- Provides major speed boosts in both training and image creation.
- Limited to single 3D images with no time variation.
- Our approach adds 4D breathing motion and phase guidance for real clinical use.

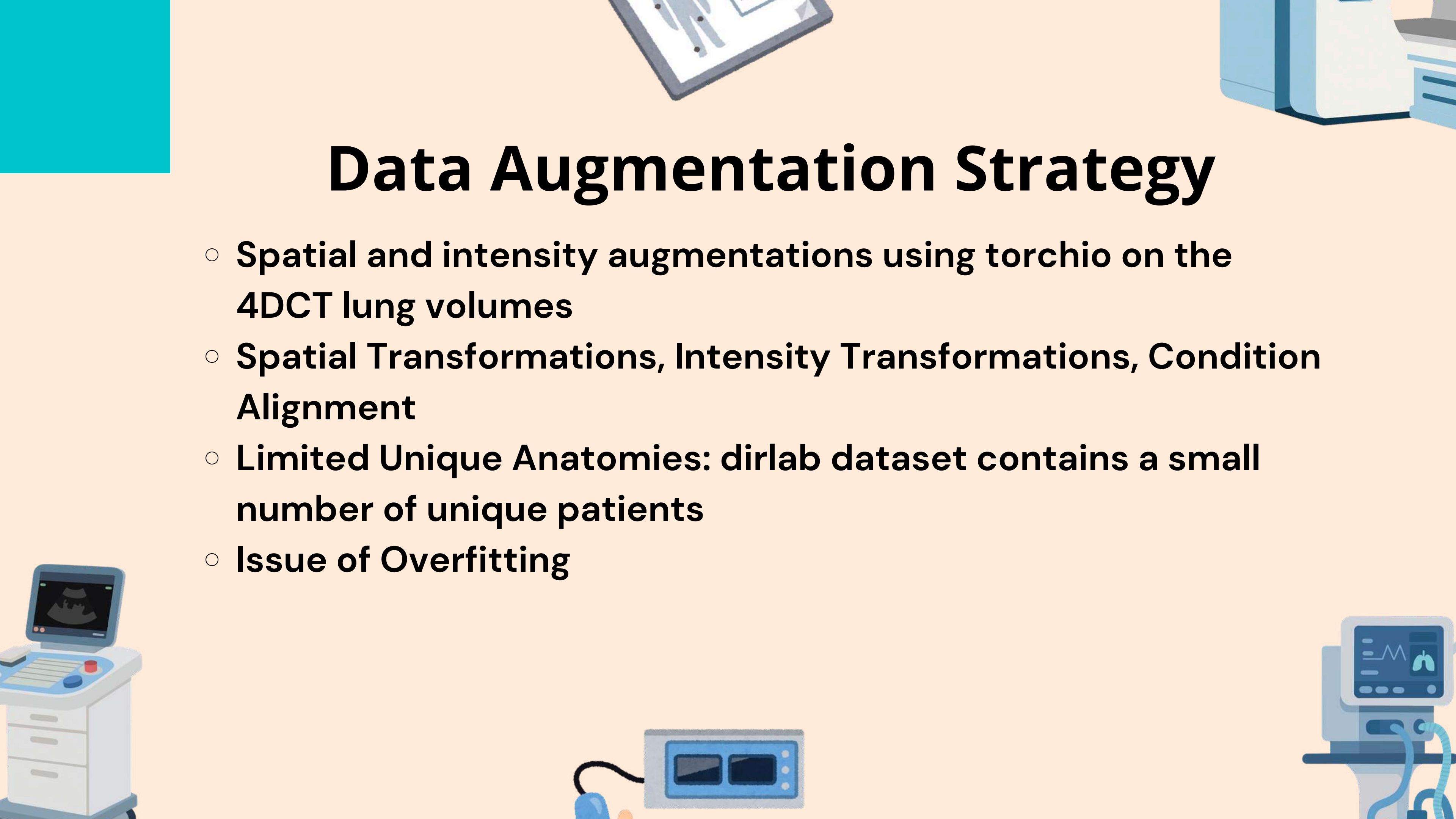




Literature Review 3

A Diffusion Model for Universal Medical Image Enhancement, by Fei, Li, Yang, Gao et al.,

- A single model, UniMIE, sharpens many kinds of medical scans with no extra training.
- It takes a noisy picture and cleans it through many tiny steps.
- It beats several older methods on CT and MRI tasks.
- It only fixes what exists, does not handle time, and stays in 2D.
- Our method fills in the missing 4D lung CT phases and checks the accuracy that matters for real anatomy.



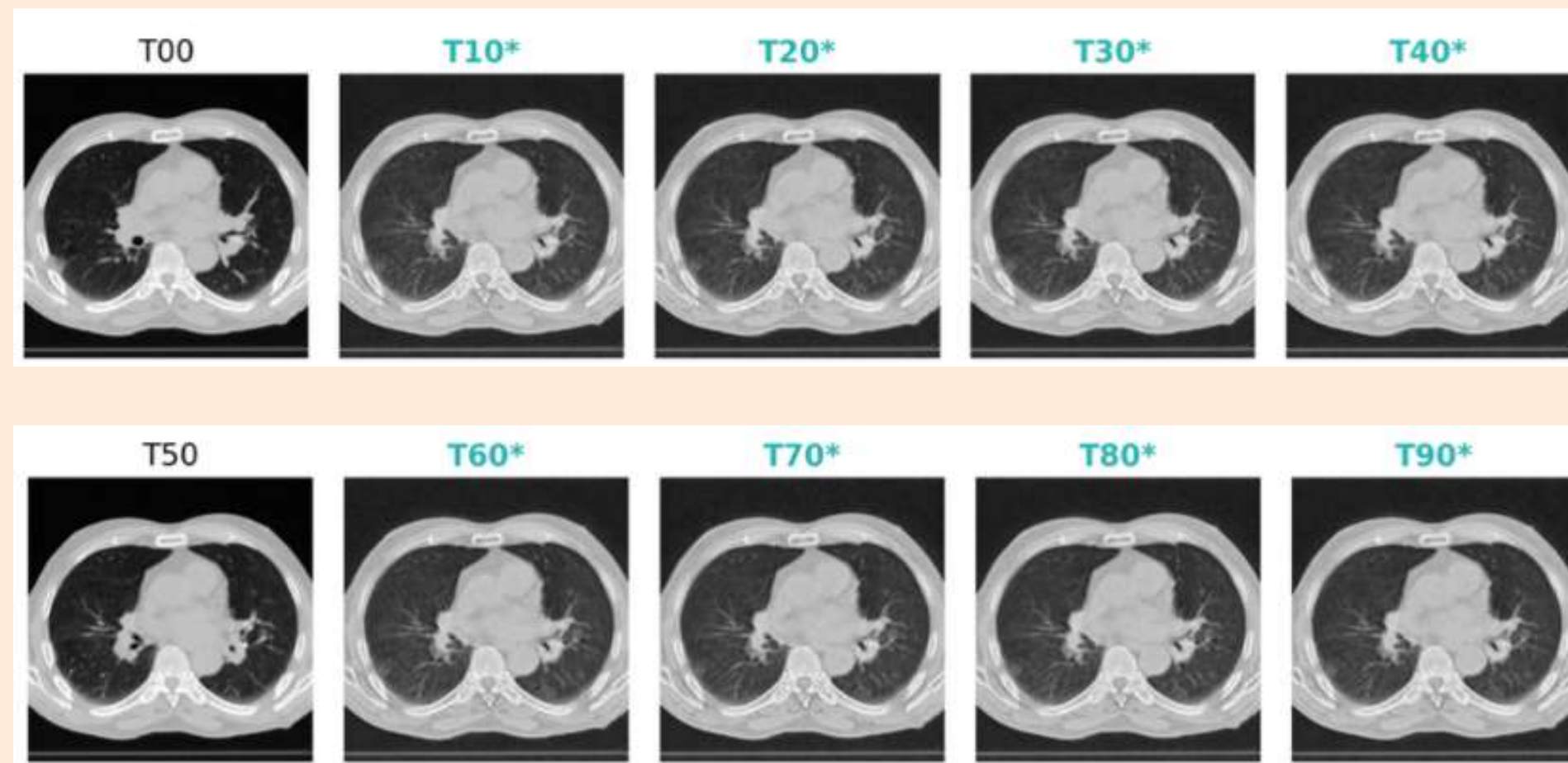
Data Augmentation Strategy

- Spatial and intensity augmentations using torchio on the 4DCT lung volumes
- Spatial Transformations, Intensity Transformations, Condition Alignment
- Limited Unique Anatomies: dirlab dataset contains a small number of unique patients
- Issue of Overfitting

Deformable Registration

Problem: only 8 real training cases to learn lung's motion

Solution: ANTsPy SyN deformable registration synthesizes additional intermediate-phase by warping DIR-Lab landmarks.



System Architecture

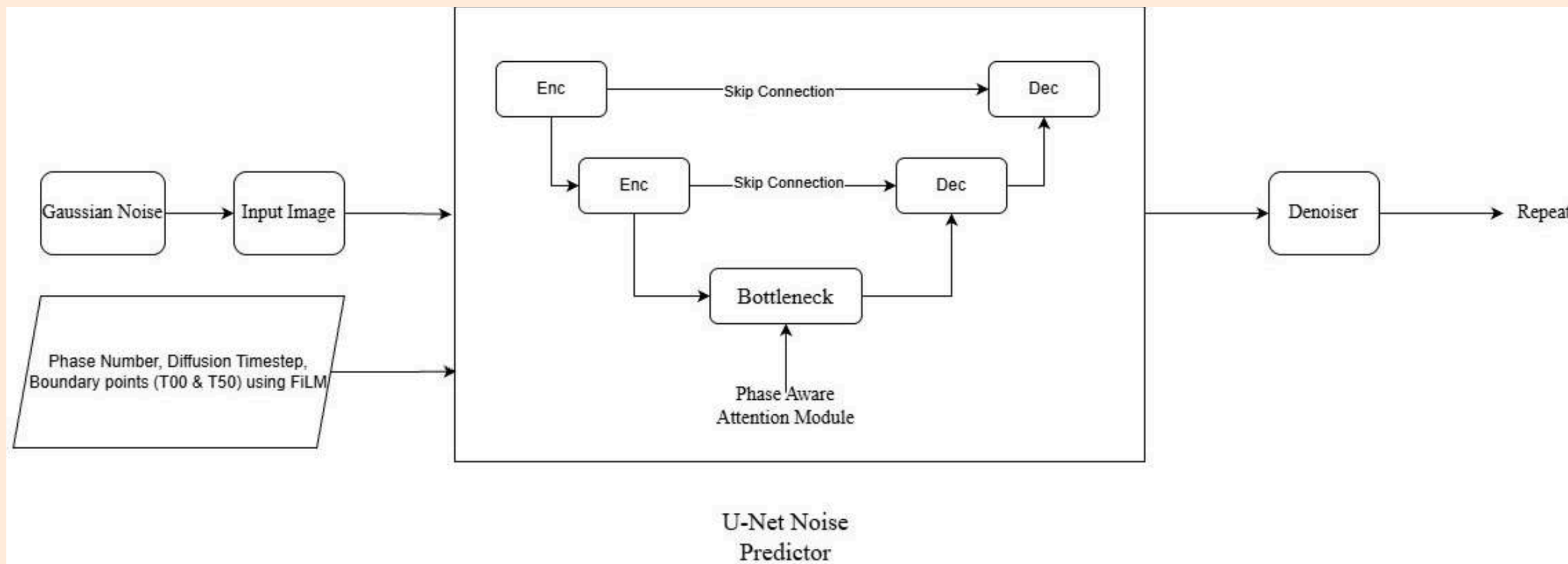
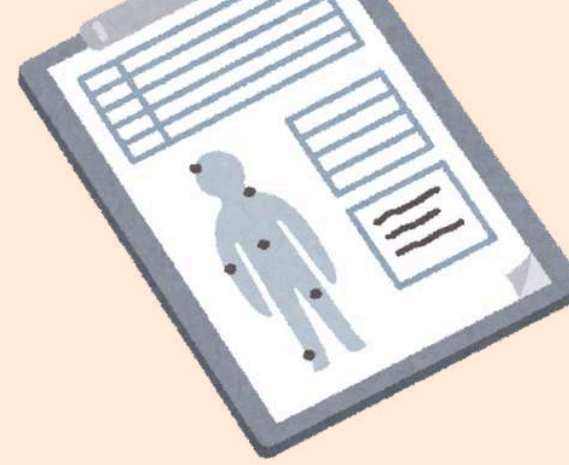
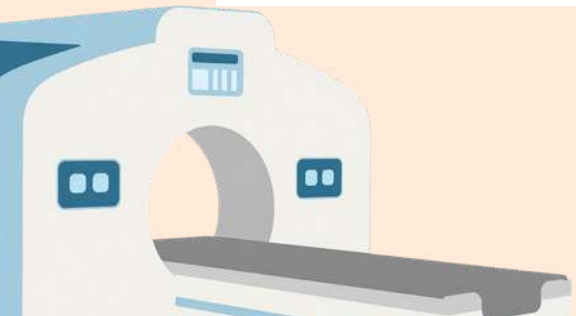


Fig: System Architecture

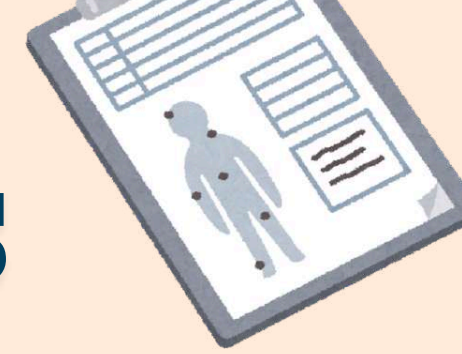
Three Conditioning Steps guide the process :

- T00 and T50 anchor points
- Phase Number
- Diffusion Timestep

Novelty:
Applying a Phase Conditioned Diffusion to an underexplored 4D CT reconstruction work.



How our model trains



1. Pick a real phase

- eg. T30 from a patient scan

2. Add Noise

- Random level t (0-1000) of Gaussian Noise

3. Feed the U-Net

- Inputs:
- [noisy T30, T00 and T50] ,
PHASE = 0.3, Timestep = t

4. Predict the noise

- Model outputs its estimate of what was added.

5. Compute the loss

- MSE between the predicted and the actual noise.

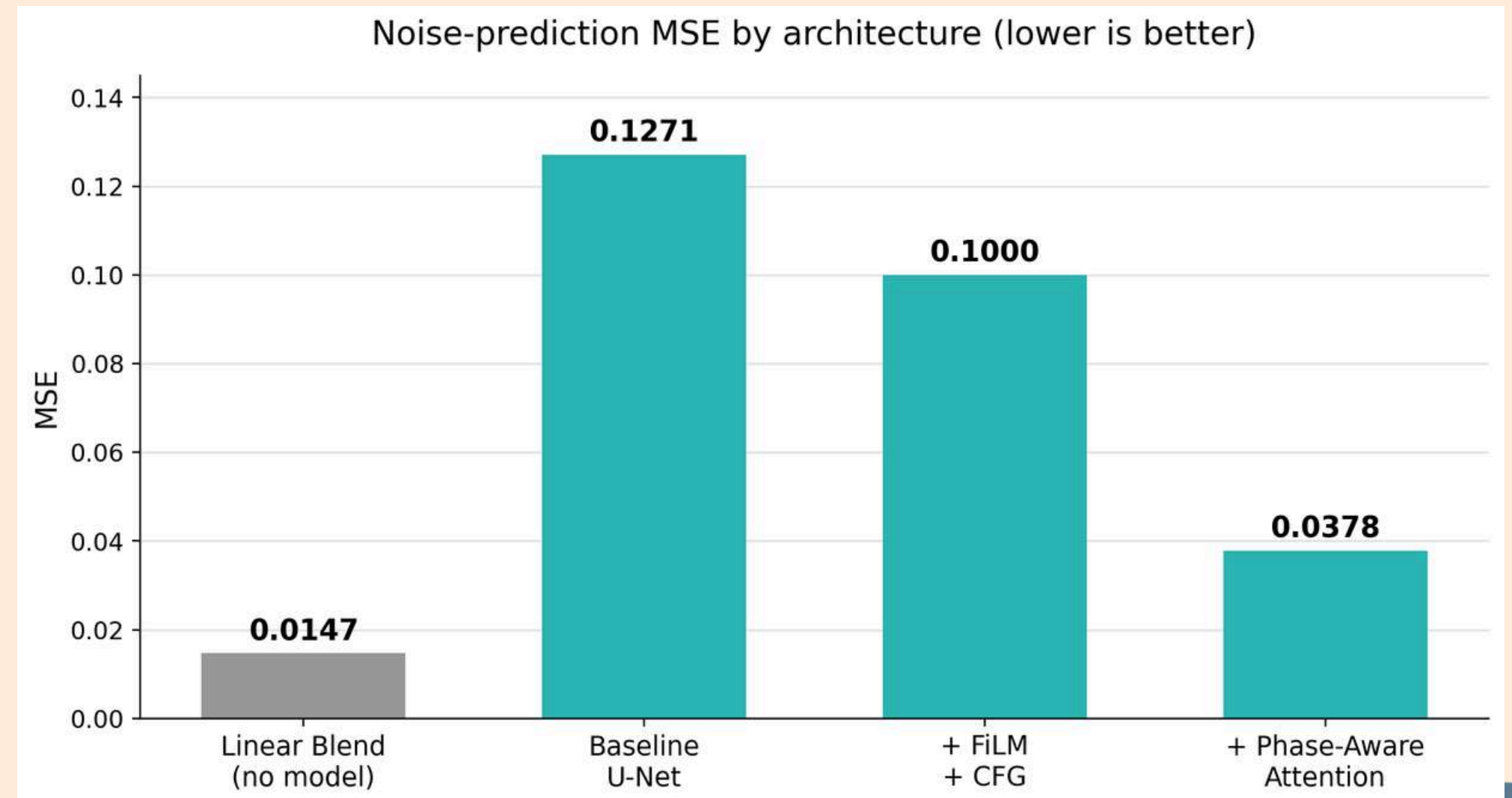
6. Backpropagate

- Repeat across all patients, phases and noise level.



MSE: Actual Metric?

- 70% upgrade from Baseline U-net to Phase Aware Attention.
- Linear Blend beat our proposed model.
- But MSE rewards blurry prediction.



What is TRE, and how do we measure it

TRE

Target Registration Error:

how far off, in millimetres, our generated volume places a landmark compared to the real DIR Lab position.

1

Cut a template

Grab a small patch around a known landmark in T00.

2

Generate the phase

The model generates that target phase.

3

Template match

Slide the patch across the volume and find the best NCC match.

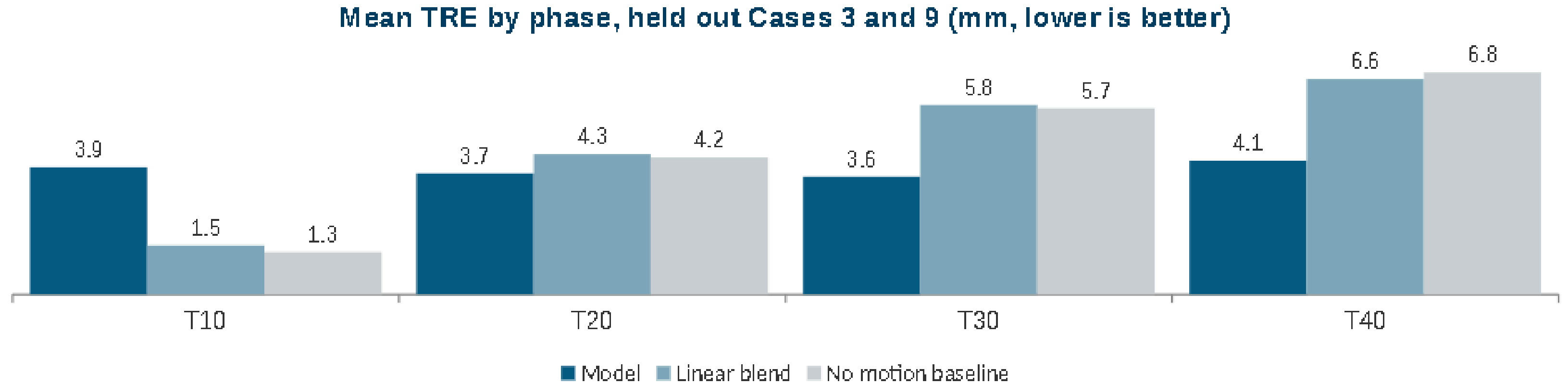
4

Measure distance

Check how far that match sits from the real DIR Lab landmark, in mm.

*Tested on **held out Cases 3 and 9**, phases T10 to T40 only, since those are the only phases with DIR Lab ground truth landmarks.*

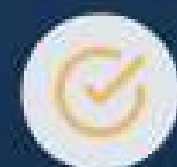
TRE results



Mean TRE: 3.85 mm for the model, vs **4.50 mm** no motion and **4.44 mm** linear blend. The model wins on average, and stays steadier across every phase.

Where the model nails it — and where it breaks down

Predicted vs. actual lung displacement, by respiratory phase



Excels at large motion

T30-T40, where the lungs actually shift — the model tracks the motion closely.



Overshoots on small motion

At T10, the lungs barely move — but the model still predicts noticeable displacement.



Real motion swings **5×** across phases $1.3 \rightarrow 6.8 \text{ mm}$

Model error barely moves $3.6 \rightarrow 4.1 \text{ mm}$

The model hasn't learned to scale its output with the respiratory phase.

Key takeaways

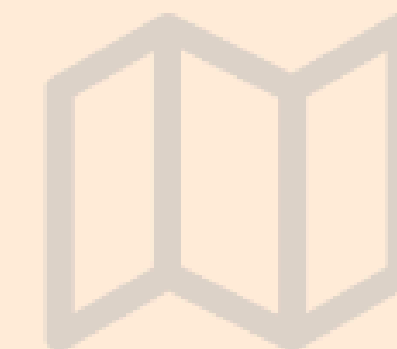
What the numbers actually mean

- 01 The model reduces landmark error by **14%** compared to assuming the lungs do not move at all.
- 02 Simple blending does not work. Averaging the two clean scans gets you **4.44 mm** — almost identical to doing nothing. Real motion isn't something you can average your way to.
- 03 The model performs best exactly where accuracy matters most — large motion phases, **T30 to T40** — when tumour position shifts the most during treatment.

This confirms the model has learned real anatomical motion — not just plausible-looking images.

Future Work

Where we take this next



Upload a web-hosted interface 01

- So more doctors, clinics, and institutions can access this technology.



Reduce model wait time 02

- Currently takes **~4 minutes** to generate the phases.
- Goal: bring generation time down to a clinically acceptable standard.



Hybrid model 03

- The model overshoots when motion is small — pair it with registration-based methods for early phases (**T10-T20**), and keep the learned model where it already wins (**T30-T40**).



Expand the training dataset 04

- Extend testing beyond held-out **Cases 3 and 9**, with more examples of small-motion phases — where the model is currently least reliable.

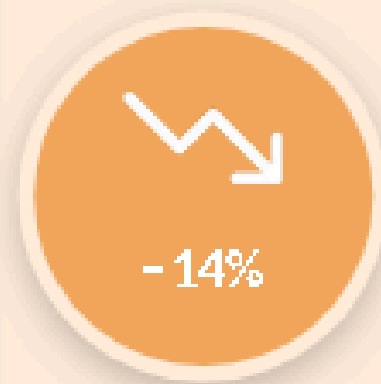
Conclusion

What we set out to prove — and where it stands



3.85 mm

OUR MODEL



4.50 mm

NO-MOTION BASELINE



The greatest limitation of our project is the **small size of the dataset**.



However, we showed the model applied in this work **beats the regular registration method**.



One step closer to better health — and a safer world.



Thank You

